Slick Trace investigation report — case_0004

Generated 2026-09-26T13:00:59.359089+00:00 · reproducible investigation report

What this document is not. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.

Finding

Leading candidate: 419210002. This is a ranking by physical consistency under tested assumptions — an investigative lead for a human analyst, not proof of responsibility.

Scene

Field	Value
Incident	case_0004
Acquired (UTC)	2026-09-10T04:30:00+00:00
Sensor	synthetic SAR-like (VV proxy, single channel)
Data status	synthetic
Pixel spacing	433.59 m
Label source	generator ground truth (synthetic) — not a human or independently confirmed label

Evidence layers

Layer	Provenance
AIS	synthetic (fictional MMSIs; no real broadcast data)
Forcing	slicktrace_core.synth_generator constant+spatial field v0.2
Licence	Synthetic data generated by this repository; released under the repository licence. No third-party data terms apply.

Traffic filtering

State	Vessels
candidate	10
excluded	1

Unassessed is reported separately from excluded: missing evidence is not exoneration, and collapsing the two would silently convert one into the other.

Rank fragility

Leader retained in 0.8375 of 80 tested perturbations. A tested perturbation changed the leader; the smallest is recorded in the JSON report.

Absence of a flip is evidence of stability across the tested set, not a proof of stability: single-axis and two-axis perturbations only.

File integrity

File	SHA-256 (first 16)
attribution_result.json	28f6266e0be58157
coverage_result.json	1e82a0203e8eba1e
forcing.npz	4b4a3d7409108fc9
forecast_result.json	360170a11e8fab91
fragility_result.json	c982850c8bf6dcf2
historical_mapping.json	0e59d782ef6b8cbb
manifest.json	26de0a4d06b4bc80
mask_groundtruth.png	744117fc89bfd972
naive_baseline_result.json	98441527f8e3dc37
next_observation_result.json	29f9f10a8c7b7dd6
scene.png	b01fa9684715b913

Hashes verify file integrity only. They do not establish that an observation is true, that a label is correct, or that this analysis is legally sufficient.

Uncertainty

• Rankings express physical consistency under tested assumptions. They are not proof of responsibility.

• Ensemble and perturbation-grid win frequencies are evidence scores, not calibrated probabilities. No calibration against held-out real outcomes exists.

• Where no candidate clears the absolute fit gate the system reports unknown source rather than naming the least-bad option.

• Synthetic ranking accuracy is not a measurement of real-world attribution accuracy.

Restated. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.